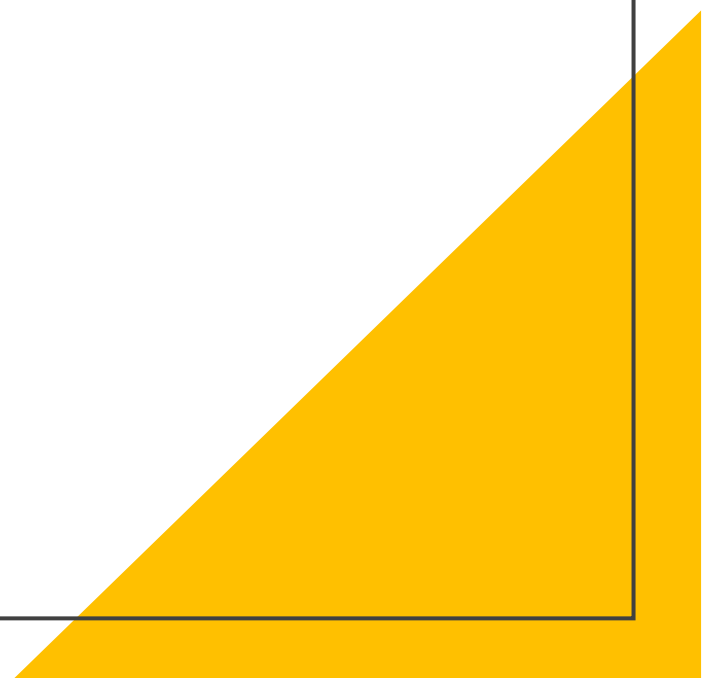


Frontend Course Session 2

HTML & first code

Created by Ahmed Samir





Agenda

- Html get started
- Html tags & Structure
- Attributes
- Links and images
- Lists
- tables
- Formatting text in html
- Iframe – audio - video
- Semantic elements
- Entities in html
- practice

HTML <p>hello world</p>

HTML (HyperText Markup Language)

In the late 1980s and early 1990s, Tim Berners-Lee, a software engineer at CERN, was trying to solve a big problem:

How can scientists around the world share and access documents easily, regardless of computer or operating system?

At that time, information was scattered, stored on different systems, and hard to link together.

[HTML history: Milestones in the web markup language](#)

[A Brief History of HTML | WIRED](#)

HTML <p>hello world</p>

How HTML Was Created

1- Chose a Markup Language: Tim Berners-Lee needed a way to structure documents on the web, inspired by SGML, a complex markup system.

2- Simplified SGML: He created a simpler version of SGML, suitable for the internet. this became HTML.

3- Designed Basic Tags: Introduced readable

4- Built First Browser & Editor: In 1990, he developed WorldWideWeb (later Nexus)

5- Wrote the First HTML Page: Hosted at CERN, explaining what the World Wide Web is and how to use it.

Released the First Specification: In 1991, he published the first HTML draft. listing 18 original tags.

Getting Started

- Open project folder in VS Code: Use `cmd + code .` (Mac) or `code .` (Windows/Linux).
- HTML file extension: Save files with `.html`
- Document Type Declaration: `<!DOCTYPE html>` — tells the browser to expect an HTML5 document.
- `html`, `head`, `title`, and `body` are required tags.
- Use `<!-- Comment -->` or `Ctrl + /` to comment.

HTML Tags

The HTML **element** is everything from the start tag to the end tag:

`<tagname>`Content goes here...`</tagname>`

Examples of some HTML elements:

`<h1>`My First Heading`</h1>`

`<p>`My first paragraph.`</p>`

Start tag	Element content	End tag
<code><h1></code>	My First Heading	<code></h1></code>
<code><p></code>	My first paragraph.	<code></p></code>
<code>
</code>	<i>none</i>	<i>none</i>

HTML Tags

Void (Empty / Self-Closing) Tags

- `<img>`
- `<br>`
- `<hr>`
- `<input>`
- `<meta>`
- `<link>`
- `<source>`

Paired Tags (Container Tags)

- `<p></p>`
- `<div></div>`
- `<h1></h1>`
- `<span></span>`
- `<a></a>`
- `<strong></strong>`

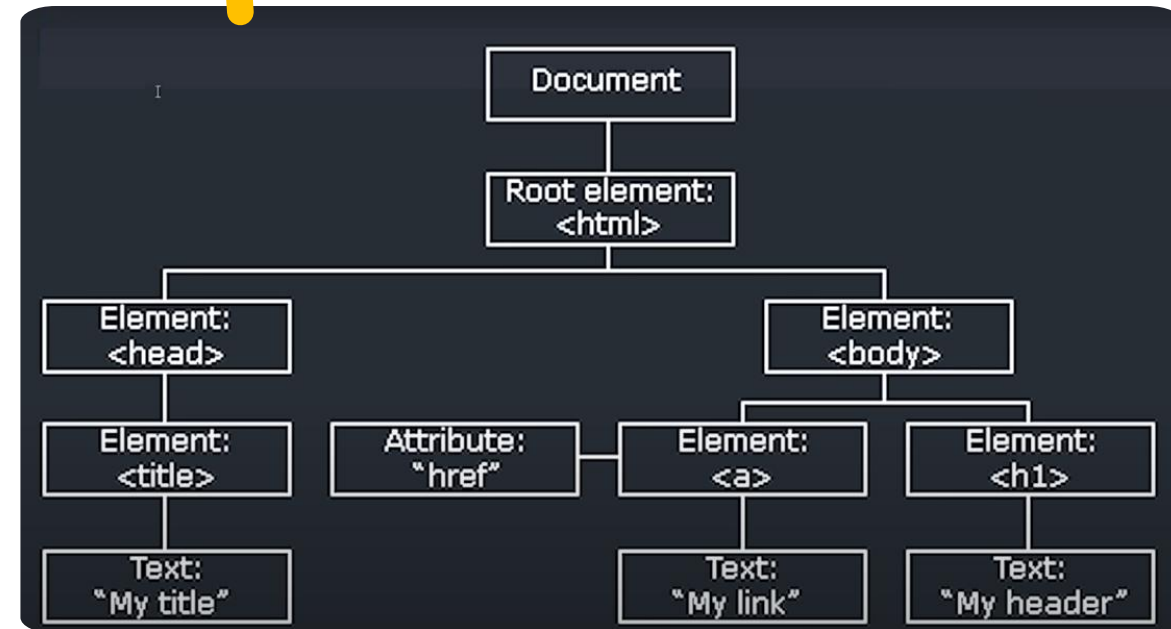
HTML Structure

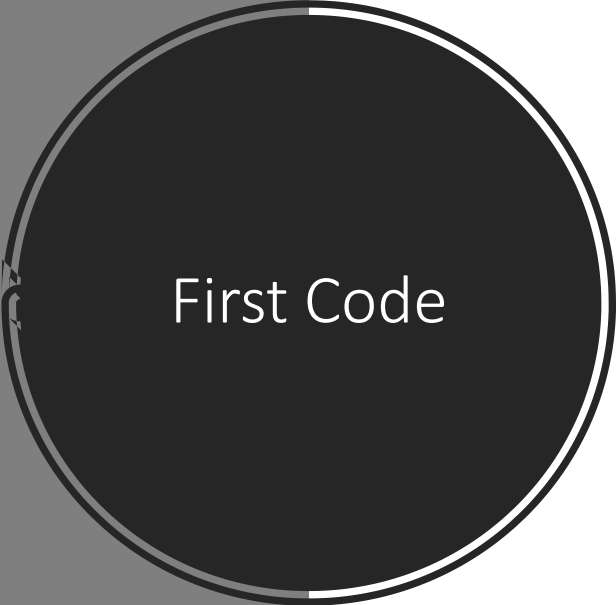
<head> Tag – The Brain of the Page

- The <head> contains metadata and non-visible information about the web page.
- It helps the browser understand how to handle the page.
- Contents inside <head> do not appear in the browser window (except indirectly).

<body> Tag – The Content of the Page

- The <body> contains everything visible to the user.
- All the text, images, buttons, forms, etc., go inside this tag.
- What you see on the screen is rendered from the <body> section.





First Code

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <h1>Welcome!</h1>
    <p>This is my first website.</p>
  </body>
</html>
```

Text Elements & Headings

- Headings: `<h1>` to `<h6>` — h1 should be used only once per page.
- Dummy text: Use lorem10 in VS Code to auto-generate text.

HTML Elements

Block Elements:

- Take full width.
- Examples: `<div>`, `<h1>`, `<p>`, `<section>`, etc.

Inline Elements:

- Appear on the same line with other elements.
- Examples: `<span>`, `<a>`, `<strong>`, etc.

Attributes

What are HTML Attributes?

Attributes provide additional information about HTML elements. They are always written in the opening tag, and usually come as name="value" pairs.



Links & Images

- Anchor Tags (<a>):

```
<a href="https://example.com" target="_blank">Visit</a>
```

target="_blank" opens the link in a new tab. Use name.html instead of ./name.html for relative internal links.

[Visit](#)

- Image Tags ():

```

```

Alt & title is important for SEO and accessibility.

Use % for responsive width.



Lists

Unordered List:

```
<ul type="circle">  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ul>
```

- Item 1
- Item 2
- Item 3

Ordered List:

```
<ol type="I" start="2">  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ol>
```

- II. Item 1
- III. Item 2
- IV. Item 3

Tables

- 1- start with <table>
 - 2- titles in <thead> or first row
 - 3- main table in <tbody>
 - 4- use <tr> to make row
 - 5- <td> for normal data cells and <th> for heading cells
- colspan — spans columns.
 - rowspan — spans rows.

Name	Job Title	Email address
Anna Fitzgerald	Staff Writer	example@company.com
John Smith	Marketing Manager	example2@company.com
Zendaya Grace	CEO	example2@company.com

teacher	day1		day2	
Ali	arabic	science	arabic2	science2
	english	french	english2	french2
mohamed	arabic	science	arabic2	science2
	english	french	english2	french2

Formatting & Layout

`<hr>` — Horizontal line (block element, empty tag).



`<br>` — Line break (not recommended when using CSS).



`<pre>` — Preserves formatting and spacing (great for code).



`<strong>`, `<small>` — Semantics and accessibility.



`<del>` — Deleted text. `<sup>` / `<sub>` — Superscript / Subscript text.

HTML Formatting Elements

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

Ifram

An <iframe> (inline frame) is an HTML tag used to embed another webpage or content (like videos, maps, or documents) within the current page.

```
<iframe src="https://www.youtube.com/embed/videoID" name="myFrame"></iframe>
```

```
<a href="page.html" target="myFrame">Open in iframe</a>
```

Video

The controls attribute adds video controls, like play, pause, and volume.

The <source> element allows you to specify alternative video files which the browser may choose from. The browser will use the first recognized format.

Add muted after autoplay to let your video start playing automatically (but muted):

```
<video controls width="500" height="300" poster="thumbnail.jpg">  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.webm" type="video/webm">  
  Your browser does not support the video tag.  
</video>
```

Audio

- The controls attribute adds audio controls, like play, pause, and volume.
- The <source> element allows you to specify alternative audio files which the browser may choose from. The browser will use the first recognized format.
- The text between the <audio> and </audio> tags will only be displayed in browsers that do not support the <audio> element.

<audio controls>

<source src="sound.mp3" type="audio/mpeg">

<source src="sound.ogg" type="audio/ogg">

Your browser does not support the audio element.

</audio>

Semantic Tags

Use instead of <div> for meaningful structure:

<nav>, <header>, <main>,
<section>, <article>, <aside>,
<footer>,<address>

```
<main>  
  <section>  
    <h3>heading</h3>  
    <article>  
      Lorem ipsum dolor sit amet consectetur adipisicing elit. Omnis quisquam dolores  
      corporis deleniti quos maxime fugit maiores, recusandae perferendis labore. Voluptates,  
      eum atque voluptatem aperiam officia laudantium eaque praesentium corrupti!  
    </article>  
  </section>  
</main>
```

HTML Entities

Reserved characters in HTML must be replaced with entities:

- Entity names look like this:

&entity_name;

- Entity numbers look like this:

&#entity_number;

[HTML Character Entities](#)

practice

Simulation you make a website for the company use :-

- 1- paragraph to describe the company
- 2- use order and unordered lists to describe services and goals
- 3- some images with alt and title attributes
- 4- links social with when I click make the tap open in a new window
- 5- put a YouTube video
- 6- use Semantic elements
- 7- make something like this Table:
- 8- make footer and write copyright with entity code

Team	Player	Statistics	
Team A	Player 1	Goals: 4	Assists: 7
	Player 2	Goals: 6	Assists: 5
Team B	Player 3	Goals: 5	Assists: 6
	Player 4	Goals: 3	Assists: 8